



University of Niagara Falls
Master Data Analytics

Principles of Analytics (DAMO-500-15)
Instructor: Mohamad Makki
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Take-Home Assignment
Student: José Alberto Martinez Morales
Student ID: NF1042389

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SECTION 1. Research Question, Objectives, and Hypotheses

1.1 Research Question

The dataset contains customer-level behavioral and demographic information used to evaluate how training interventions and promotional strategies influence engagement, spending, satisfaction, and conversion outcomes. Based on this information, we can formulate the following research question:

“Do customers assigned to different training groups exhibit significantly different levels of engagement?”

This question is clear, measurable, and suitable for inferential analysis using ANOVA or non-parametric alternatives.

1.2 Variables Identification

Dependent Variable

- **Engagement_Minutes**

Represents the total number of minutes a customer interacted with the platform. This is the primary outcome used to compare training group effectiveness.

Independent Variable

The categorical condition expected to influence engagement:

- **Training_Group**

(Treatment_A, Treatment_B, Treatment_C)

This variable assigns customers to different training interventions.

Control Variables

These demographic and economic factors may influence engagement and are acknowledged to reduce confounding:

- **Age**
- **Annual_Income**

Additional variable that are not part of the primary research question but available for extended analysis.

Promotion_Exposure, Gender, Region, Department, Monthly_Spend, Satisfaction_1to10, Performance_Score, Purchase_Converted

These variables are not required for the primary test but may support secondary insights.

1.3 Hypotheses

Based on the variables and the research question, the following hypothesis are proposed to examine differences, associations, probabilities, and group comparisons within the dataset.

Difference Hypothesis (t-test)

H₀: There is **not statistically significant difference** in mean engagement minutes between customers exposed and not exposed to the promotion.

H_A: There is a **statistically significant difference** in mean engagement minutes between customers exposed and not exposed to the promotion.

Association Hypothesis (Correlation)

H₀: Monthly spend is **not statistically significantly correlated** with satisfaction scores.

H_A: Monthly spend is **statistically significantly correlated** with satisfaction scores.

Probability-Based Hypothesis (Proportion Test)

H_0: The probability of purchase conversion **does not differ statistically** between customers in the training group and those not in the training group.

H_A: The probability of purchase conversion **differs statistically** between customers in the training group and those not in the training group.

ANOVA-Based Hypothesis

H_0: Mean monthly spend **does not differ statistically** across regions.

H_A: At least one region shows **a statistically significant difference** in mean monthly spend.

SECTION 2. Data Preparation and & Screening

2.1 Initial Data Screening

Before conducting outlier detection, the dataset was screened for missing values and duplicate records using SPSS.

Output A1 shows that the dataset contains **no missing values**, as all variables report complete cases with zero system-missing entries.

Output A2 confirms that there are **no duplicate rows**, indicating that the dataset does not require row-level cleaning, imputation, or removal of repeated observations.

These results demonstrate that the dataset is structurally complete and ready for outlier analysis.

2.2 Outlier Detection

This subsection shows the results of the outlier screening using descriptive statistics, the IQR method, and Z-scores. **Output A3** presents the descriptive statistics (Mean, SD, Min, Q1, Median, Q3, Max) before any screening. **Output A4** shows the IQR outlier analysis for all continuous variables. Among these, Engagement_Minutes (7.5 %) and Monthly_Spend (5.83 %) exceed the 5% threshold, while the remaining variables fall below it (See **Calculation C1** for details) **Output A5** displays the Z-scores results for all six continuous variables, with each variable showing fewer than 5% of cases beyond ± 3 SD threshold (See **Calculation C2** for details). Since both the IQR method and Z-score method indicate that the extreme values reflect legitimate customer behavior and fall within acceptable analytical limits, **all outliers were retained**. **Output A6** (After table) is identical to **Output A3** because no observations were removed.

SECTION 3. Descriptive Statistics & Visualizations

3.1 Descriptive Statistics

Output A7 Shows the central tendency, variability, and distribution shape for the four key variables. Engagement and spending are highly right skewed with extreme upper values, while satisfaction and performance remain more balanced. These patterns highlight strong behavioral variability across customers.

3.2 Visualizations

Figure B1 Bar Chart of customer gender distribution.

The sample shows a relatively balanced gender distribution, with slightly more male customers (215) compared to female (187) and other identities (198). This balance suggests no major gender bias in the dataset.

Figure B2 Histogram of Monthly Spend.

Monthly spending is strongly right skewed, with most customers spending under \$2,000 and a small number of high spending customers creating a long upper tail.

Figure B3 Scatterplot of Engagement Minutes vs. Performance Score.

Most observations cluster between 0-200 engagement minutes and performance scores of 60-90, with a few high-engagement outliers. The pattern suggests a weak positive relationship: higher engagement is associated with slightly higher performance, but the trend is not strongly linear.

Figure B4 Boxplot of Monthly Spend by Department.

The distribution of monthly spending varies noticeably across departments. Retail shows moderate spending with fewer extreme values, online displays the widest spread and several high-spend outliers, and B2B exhibits the highest median spending overall. These differences suggest that customer spending behavior is strongly influenced by departmental context.

SECTION 4. Probability & Association Analysis

4.1 Probability Questions

Normal Distribution

What is the probability that a randomly selected customer has a monthly level exceeding \$2,000, assuming Monthly_Spend follows a normally approximated distribution based on the sample mean and variance?

All derivations for this problem are presented in **Calculation C3**.

Interpretation.

Approximately **14.6%** of customers spend more than \$2,000 per month. This aligns with the right skewed distribution where a minority of high spending customers drive the upper tail.

Binomial Probability

What is the probability that exactly four out ten randomly selected customers belong to Retail department, assuming independent selections with success probability equal to the empirical Retail proportion in the dataset?

All derivations for this problem are presented in **Calculation C4**.

Interpretation.

The probability of **22.9%** indicates that getting exactly four Retail customers in a group of ten is a reasonably common outcome.

4.2 Association Analysis

Correlation Analysis: Engagement Minutes vs. Performance Score

Interpretation.

Figure 3.3 shows a slight upward trend, and the normality checks in **Output A8** confirms violations, so Spearman was appropriate. **Output A9** reports the test statistic, p-value, and effect

size ($p = 0.1337$), indicating a weak but statistically significant positive association. Overall, engagement has only a small influence on performance despite meeting the assumptions for Spearman. (Numiqo, Spearman Rank Correlation — Simply Explained , (2021))

Association Analysis: Promotion Exposure vs. Purchase Conversion.

Assumption Checks

- **Expected counts:** All expected frequencies were ≥ 5 (See **Output A10**), so the Chi-square assumptions were satisfied.
- **Independence:** Each row represents a unique customer.

Test Results

- **Chi-square statistic (X^2):** 4.015
- **P-value:** 0.0451
- **Effect size (Cramer's V):** 0.0818 (small effect)

Interpretation.

The Chi-square test shows a statistically significant association between Promotion Exposure and Purchase Conversion ($X^2 = 4.015$, $p = 0.045$), with assumptions confirmed by the expected-count check. Cramer's V (0.0818) indicates that although the relationship is significant, the effect size is **small**. Overall, exposure to a promotion is related to conversion behavior, but the practical impact of this association is limited (Numiqo, Chi2 Test – Easily Explained with an Example, 2021)

SECTION 5. ANOVA & Non-Parametric Comparison

Before conducting the inferential analysis, a series of diagnostic checks were performed to evaluate whether the assumptions required for a one-way ANOVA were satisfied. These included assessing the normality of the Monthly Spend distributions within each department, testing the equality of variances across groups, and examining the normality of the model residuals. The results of these assumption checks (**Output A11-A13**) guided the decision on whether ANOVA was appropriate or whether a non-parametric alternative, such as the Kruskal-Wallis's test, was required.

5.1 Assumption Checks

Normality of Within-Group Distributions

Shapiro-Wilk tests were conducted separately for Retail, Online, and B2B. All three departments showed extreme deviations from normality (Retail $p < 0.001$; Online $p < 0.001$; B2B $p < 0.001$), as shown in **Output A11**. These results align with the strong right skewness and numerous outliers visible in the boxplot (**Figure 3.4**), indicating that the normality assumption for ANOVA is not satisfied.

Levene's Test for Homogeneity of Variances

Levene's test (**Output A12**) showed no significant difference in variances across the three departments ($F = 2.888$, $p = 0.056$). This suggests that the homogeneity of variance assumption is met.

Normality of Residuals

Residuals from the ANOVA model were tested using Shapiro-Wilk. The result (statistic = 0.793, $p = 0.001$), shown in **Output A13**, indicates extreme non-normality. This confirms that the ANOVA assumption of normally distributed is violated even after accounting for group affects.

Assumption Summary

Although variances are equal, the severe non-normality of both the raw group distributions and the model residuals indicates that ANOVA assumptions are not plausible. Therefore, the **Kruskal-Wallis's test** is the appropriate inferential method for comparing Monthly Spend across departments.

5.2 Inferential Test

ANOVA

The one-way ANOVA results are presented in **Output A14** and summarizes in the following table.

Source	sum_sq	df	F	p-value
Department	1.0467e+07	2	7.083	0.000912
Residual	4.4112e+08	597	—	—

Although ANOVA indicates significant differences in Monthly Spend across departments, **this result is unreliable due to violated assumptions**. (Numiqo, ANOVA – A Full Lecture to Learn Analysis of Variance , 2023)

Kruskal-Wallis's Test

The Kruskal-Wallis's test results (**Output A15**) are:

- **H-statistic:** 19.097
- **p-value:** 0.0000713
- **Effect size (epsilon-squared):** 0.029 (small effect)

The Kruskal-Wallis's test confirms a statistically significant difference in Monthly Spend across the three departments.

5.3 Interpretation

The Kruskal-Wallis's test evaluates whether the **distributions** of Monthly Spend differ across groups, rather than comparing means as in ANOVA. The results show that Monthly Spend differs significantly across Retail, Online, and B2B departments, indicating that at least one group has a distinct spending distribution. Although the effect size is small ($\epsilon^2 = \mathbf{0.029}$), the difference is still meaningful in practical terms. Because the test is significant, a post-hoc procedure such as Dunn's test would be required to determine which specific departments differ, although this is not required for the assignment. Overall, the results suggest that department membership influences customer spending behavior, but the magnitude of these differences is modest.

(Numiqo, Kruskal–Wallis Test (Simply Explained), 2020)

For an extended explanation of the Kruskal–Wallis's test, including optional Python simulations and visualizations, see Appendix D. This material is included for illustration and is not part of the graded analysis.

References

Numiqo. ((2021)). *Spearman Rank Correlation — Simply Explained* . Retrieved from YouTube

video: https://www.youtube.com/watch?v=XV_W1w4Nwoc

Numiqo. (2020). *Kruskal–Wallis Test (Simply Explained)*. Retrieved from YouTube video:

<https://www.youtube.com/watch?v=l86wEhUzkY4>

Numiqo. (2021). *Chi2 Test – Easily Explained with an Example*. Retrieved from YouTube video:

<https://www.youtube.com/watch?v=EjtXk-yEK6w>

Numiqo. (2023). *ANOVA – A Full Lecture to Learn Analysis of Variance* . Retrieved from

YouTube video: <https://www.youtube.com/watch?v=wraov1Nv7UA>

Appendix A SPSS output

Output A1 Descriptive Statistics and Missing Values Check

Descriptive Statistics					
	N	Minimum	Maximum	Mean	Std. Deviation
Purchase_Converted	600	0	1	.33	.471
Age	600	18	70	39.20	11.509
Annual_Income	600	18000	250000	68659.83	38883.256
Engagement_Minutes	600	19.5	1200.0	146.578	113.5493
Monthly_Spend	600	139.25	7432.88	1083.8617	868.27937
Satisfaction_1to10	600	2.3	10.0	6.719	1.5098
Performance_Score	600	33.1	100.0	77.119	9.3462
Returns_Count	600	0	16	3.29	2.430
Valid N (listwise)	600				

Output A2 Duplicate Values Check (No duplicates Detected)

Statistics

Indicator of each first matching c

N	Valid	600
	Missing	0

Indicator of each first matching case as Primary

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Primary Case	600	100.0	100.0	100.0

Output A3 Descriptive Statistics including mean, standard deviation, and five-number summary

Descriptive Statistics							
	N Statistic	Minimum Statistic	Maximum Statistic	Mean Statistic	Std. Deviation Statistic	Skewness Statistic	Std. Error
Age	600	18	70	39.20	11.509	.099	.100
Annual_Income	600	18000	250000	68659.83	38883.256	1.346	.100
Engagement_Minutes	600	19.5	1200.0	146.578	113.5493	3.182	.100
Monthly_Spend	600	139.25	7432.88	1083.8617	868.27937	2.414	.100
Satisfaction_1to10	600	2.3	10.0	6.719	1.5098	-.196	.100
Performance_Score	600	33.1	100.0	77.119	9.3462	-.400	.100
Valid N (listwise)	600						

Percentiles

		Percentiles		
		25	50	75
Weighted Average (Definition 1)	Age	31.25	39.00	47.00
	Annual_Income	40193.00	58897.00	85095.25
	Engagement_Minutes	78.425	117.250	168.200
	Monthly_Spend	531.0650	817.8800	1375.9075
	Satisfaction_1to10	5.700	6.800	7.800
	Performance_Score	71.000	77.500	83.725
Tukey's Hinges	Age	31.50	39.00	47.00
	Annual_Income	40244.00	58897.00	85030.50
	Engagement_Minutes	78.650	117.250	168.100
	Monthly_Spend	531.7700	817.8800	1375.7050
	Satisfaction_1to10	5.700	6.800	7.800
	Performance_Score	71.000	77.500	83.650

Output A4 IQR Outlier Analysis

See **Calculation C1** for a clear explanation of the table.

Variable	IQR	Lower_Bound	Upper_Bound	Outlier_ %	Decision
Age	15.2500	8.87500	69.87500	0.33	Keep (outliers < 5%)
Annual_Income	44670.7500	-26711.12500	151971.87500	5.00	Consider removal (> 5%)
Engagement_Minutes	89.1250	-54.81250	301.68750	7.50	Consider removal (> 5%)
Monthly_Spend	843.0275	-732.06625	2640.04375	5.83	Consider removal (> 5%)

Satisfaction_1to10	2.1000	2.55000	10.95000	0.67	Keep (outliers < 5%)
Performance_Score	12.5750	52.13750	102.43750	0.67	Keep (outliers < 5%)

Output A5 Z-score Outlier Detection

See **Calculation C2** for a clear explanation of the table.

Variable	Lower_Z	Upper_Z	Outlier_%	Decision
Age	-3	3	0.00	Keep (outliers < 5%)
Annual_Income	-3	3	1.33	Keep (outliers < 5%)
Engagement_Minutes	-3	3	2.17	Keep (outliers < 5%)
Monthly_Spend	-3	3	2.17	Keep (outliers < 5%)
Satisfaction_1to10	-3	3	0.00	Keep (outliers < 5%)
Performance_Score	-3	3	0.67	Keep (outliers < 5%)

Output A7 Descriptive Statistics for Four Key Variables

Descriptive Statistics									
	N Statistic	Minimum Statistic	Maximum Statistic	Mean Statistic	Std. Deviation Statistic	Skewness		Kurtosis	
Engagement_Minutes	600	19.5	1200.0	146.578	113.5493	3.182	.100	17.243	.199
Monthly_Spend	600	139.25	7432.88	1083.8617	868.27937	2.414	.100	8.963	.199
Satisfaction_1to10	600	2.3	10.0	6.719	1.5098	-.196	.100	-.198	.199
Performance_Score	600	33.1	100.0	77.119	9.3462	-.400	.100	1.161	.199
Valid N (listwise)	600								

Output A8 Normality Check

To assess whether *Engagement Minutes* and *Performance Score* follow a normal distribution, the Kolmogorov-Smirnov and Shapiro-Wilk tests were examined in SPSS.

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Engagement_Minutes	.183	600	<.001	.730	600	<.001
Performance_Score	.028	600	.200*	.986	600	<.001

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Engagement Minutes

- **Kolmogorov-Smirnov:** $p < 0.001$
- **Shapiro-Wilk:** $p < 0.001$
- **Interpretation:** Both tests are significant, indicating a clear violation of normality.

Performance Score

- **Kolmogorov-Smirnov:** $p = 0.200$ (lower bound)
- **Shapiro-Wilk:** $p < 0.001$

- **Interpretation:** Although the Kolmogorov-Smirnov test appears non-significant, the Shapiro-Wilk test (more reliable for $n < 2000$) is significant. This confirms that Performance Score also violates the assumption of normality.

Both variables violate the assumption of normality. Because of this, a **non-parametric correlation test (Spearman's rho)** is more appropriate than Pearson's correlation for analyzing the relationship between Engagement Minutes and Performance Score.

Output A9 Spearman's Rank Correlation

Because the normality assumption was violated for both variables, a **Spearman's rank-order correlation** was conducted to examine the relationship between *Engagement Minutes* and *Performance Score*.

Correlations			Engagement_Minutes	Performance_Score
Spearman's rho	Engagement_Minutes	Correlation Coefficient	1.000	.134**
		Sig. (2-tailed)	.	.001
		N	600	600
	Performance_Score	Correlation Coefficient	.134**	1.000
		Sig. (2-tailed)	.001	.
		N	600	600

** . Correlation is significant at the 0.01 level (2-tailed).

Output A9 shows a **weak but statistically significant positive association** between the two variables:

- **Spearman's rho (ρ): 0.134**
- **P = 0.001 (2-tailed)**
- **N=600**

The positive coefficient indicates that higher engagement is associated with slightly higher performance scores. However, the effect size is small, suggesting that while the relationship is statistically significant, it is not practically strong.

Output A10 : Association Analysis Promotion Exposure vs. Purchase Conversion

The Chi-square test indicates a **statistically significant association** between Promotion Exposure and Purchase Conversion ($X^2 = 4.015$, $p = 0.045$). The expected-counts assumption was satisfied, confirming the validity of the test. Although the association is statistically significant, the effect size is **small** (Cramer's $V = 0.085$), suggesting that promotion exposure influences conversion behavior, but practical impact is limited. _

Chi-Square Tests					
	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	4.375 ^a	1	.036		
Continuity Correction ^b	4.015	1	.045		
Likelihood Ratio	4.412	1	.036		
Fisher's Exact Test				.043	.022
N of Valid Cases	600				

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 83.91.

b. Computed only for a 2x2 table

Symmetric Measures				
		Value	Approximate Significance	
Nominal by Nominal	Phi	.085	.036	
	Cramer's V	.085	.036	
	Contingency Coefficient	.085	.036	
N of Valid Cases		600		

Output A11: Normality Check for ANOVA

To assess whether the ANOVA normality assumption was satisfied, **Shapiro-Wilk tests were conducted separately for each Department group** (B2B, Online, and Retail) using Monthly Spend as the dependent variable.

Tests of Normality							
	Department	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
Monthly_Spend	B2B	.164	182	<.001	.761	182	<.001
	Online	.164	211	<.001	.821	211	<.001
	Retail	.179	207	<.001	.793	207	<.001

a. Lilliefors Significance Correction

All three departments showed **extreme deviations from normality**, as indicated by highly significant Shapiro-Wilk p-values:

- **B2B:** $W = 0.761$, $p < 0.001$
- **Online:** $W = 0.821$, $p < 0.001$
- **Retail:** $W = 0.793$, $p < 0.001$

These results confirm that **none of the groups follow a normal distribution**. This supports the decision to use a non-parametric alternative (**Kruskal-Wallis**) for comparing Monthly Spend across departments.

Output A12: Homogeneity of Variance

To assess whether the ANOVA assumption of equal variances was satisfied, **Levene's test for Homogeneity of Variance** was conducted for Monthly Spend across the three departments, (B2B, Online, Retail).

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Monthly_Spend	Based on Mean	2.888	2	597	.056
	Based on Median	2.185	2	597	.113
	Based on Median and with adjusted df	2.185	2	548.639	.113
	Based on trimmed mean	2.421	2	597	.090

SPSS reports four versions of Levene’s test. The standard version used for ANOVA is “based on mean”.

- **Levene statistics:** 2.888
- **Df1:** 2
- **Df2:** 597
- **P = 0.056**

Because the p-value for Levene’s Test (based on mean) is **greater than 0.05**, the assumption of **homogeneity of variances is satisfied**. This indicates that the variability in Monthly Spend is statistically similar across the Retail, Online, and B2B departments.

Output A13: ANOVA and Residual

To evaluate whether Monthly Spend differed across the three departments (Retail, Online, B2B), a one-way ANOVA model was estimated in SPSS. As part of the diagnostic process, **residuals were saved** to assess the normality assumption.

The residuals (RES_1) were generated in SPSS by running the ANOVA model and saving the **unstandardized residuals**.

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Residual for Monthly_Spend	.155	600	<.001	.793	600	<.001

a. Lilliefors Significance Correction

Although the ANOVA table provides an F-statistic and p-value, the earlier assumption checks (Output A11 and Output A13) showed:

- **Severe non-normality** in all groups
- **Residuals also non-normal ($p < 0.001$)**

Because ANOVA assumptions are violated, the ANOVA results are **not valid for inference**, and the correct test is the **Kruskal-Wallis' test**.

Output A14: ANOVA

The one-way ANOVA comparing **Monthly Spend** across the three departments indicated a **statistically significant difference** in mean.

ANOVA					
Monthly_Spend					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	10467282.663	2	5233641.332	7.083	<.001
Within Groups	441124248.56	597	738901.589		
Total	451591531.23	599			

Because the **normality assumption is not met**, the ANOVA results **cannot be used for valid inference**, and a **non-parametric test (Kruskal-Wallis)** is required to confirm whether the differences between departments are statistically meaningful.

Output A15: Kruskal-Wallis's test

Because the assumptions for ANOVA were violated (non-normality of groups and residuals), a Kruskal-Wallis' test was conducted to compare the Monthly Spend across the three departments. The test showed a statistically significant difference in spending levels, $H(2) = 19.097$, $p < 0.001$, indicating that at least one department differs from the others. The effect size was calculated here using the data from the tables, epsilon-squared, was $\epsilon^2 = 0.029$, representing a small effect.

Ranks			
	Dep_code	N	Mean Rank
Monthly_Spend	1.00	207	264.99
	2.00	211	299.58
	3.00	182	341.95
	Total	600	

Test Statistics^{a,b}

Monthly_Spend	
Kruskal-Wallis H	19.097
df	2
Asymp. Sig.	<.001

a. Kruskal Wallis Test

b. Grouping Variable:
Dep_code

Calculating the effect size (epsilon-square)

Formula

$$\epsilon^2 = \frac{H - k + 1}{N - k}$$

Where:

- $H = 19.097$
- $K = 3$ groups
- $N = 600$ total cases

Calculation

$$\epsilon^2 = \frac{19.097 - 3 + 1}{600 - 3}$$

$$\epsilon^2 = \frac{17.097}{597} = 0.0286$$

Effect size

$$\epsilon^2 = 0.029 \text{ (small effect)}$$

Appendix B Visualizations

Figure B1

The dataset shows a relatively balanced distribution across gender categories, with slightly more male customers, followed by other gender identities and female customers. This balance suggests no major gender bias in the sample.

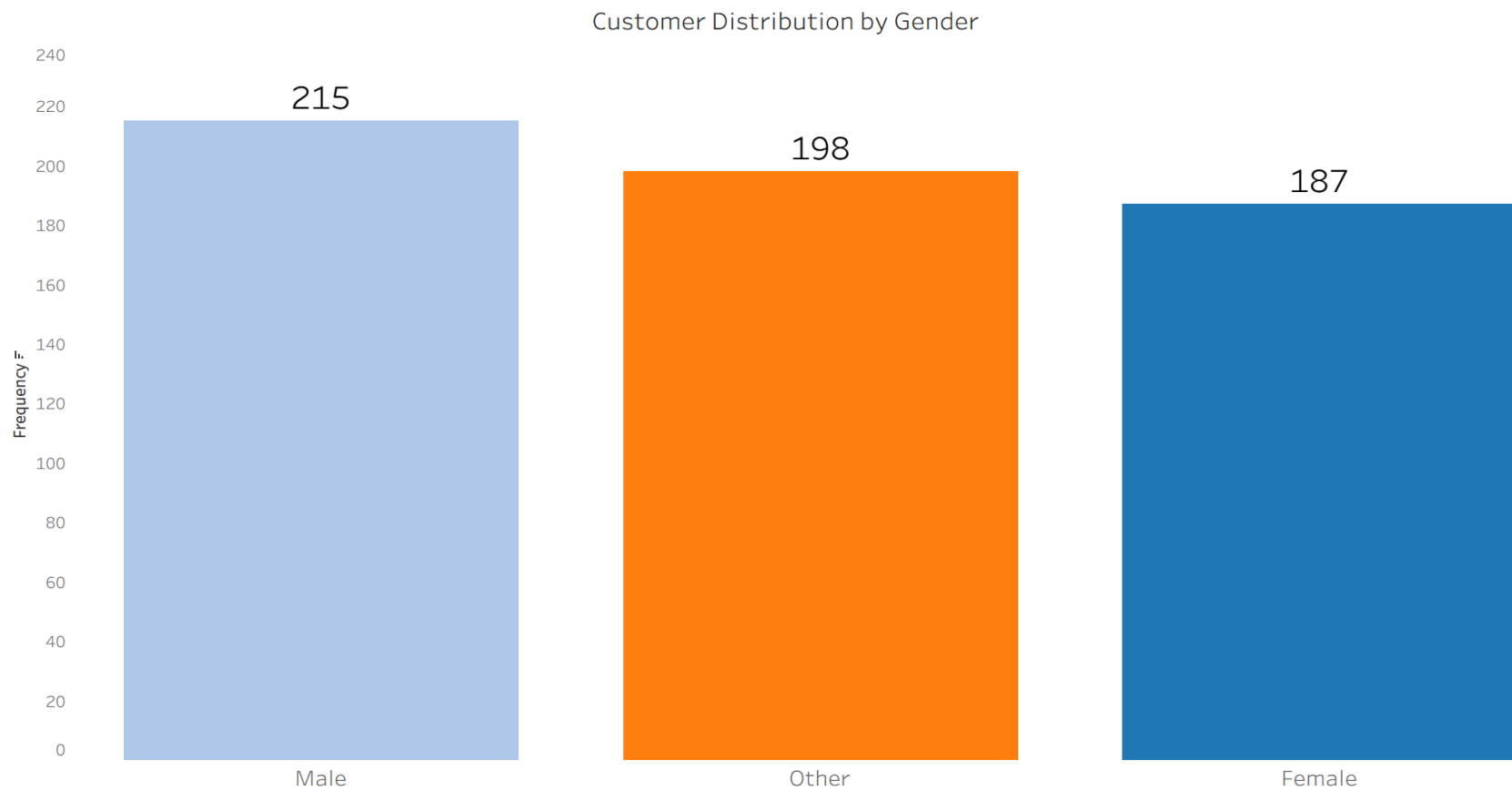


Figure B2

Monthly spending is right-skewed, with most customers spending at lower levels and a smaller group of high-spending customers creating a long upper tail.

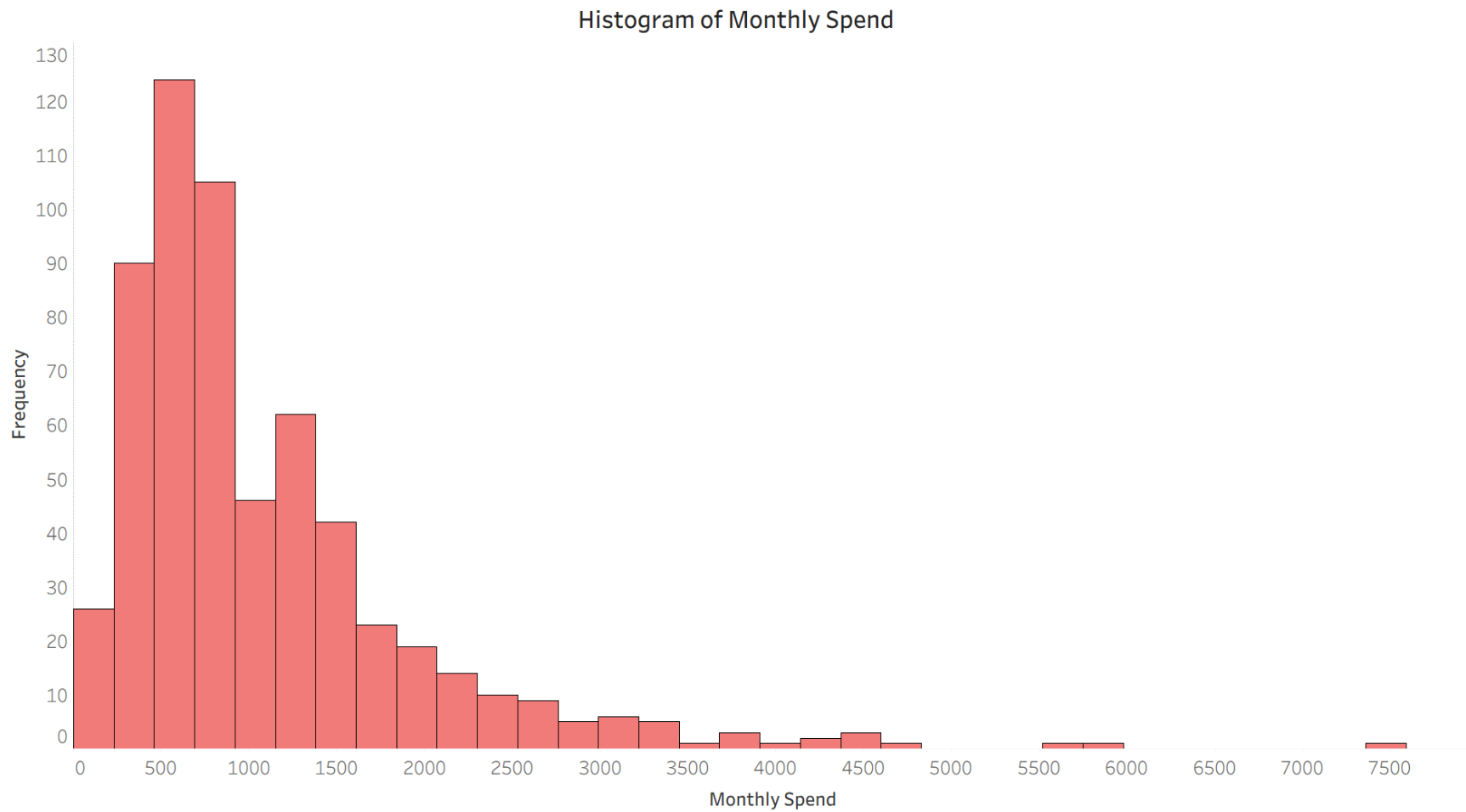


Figure B3

The scatterplot reveals a weak positive association, where higher engagement aligns with slightly higher performance, consistent with the Spearman correlation results.

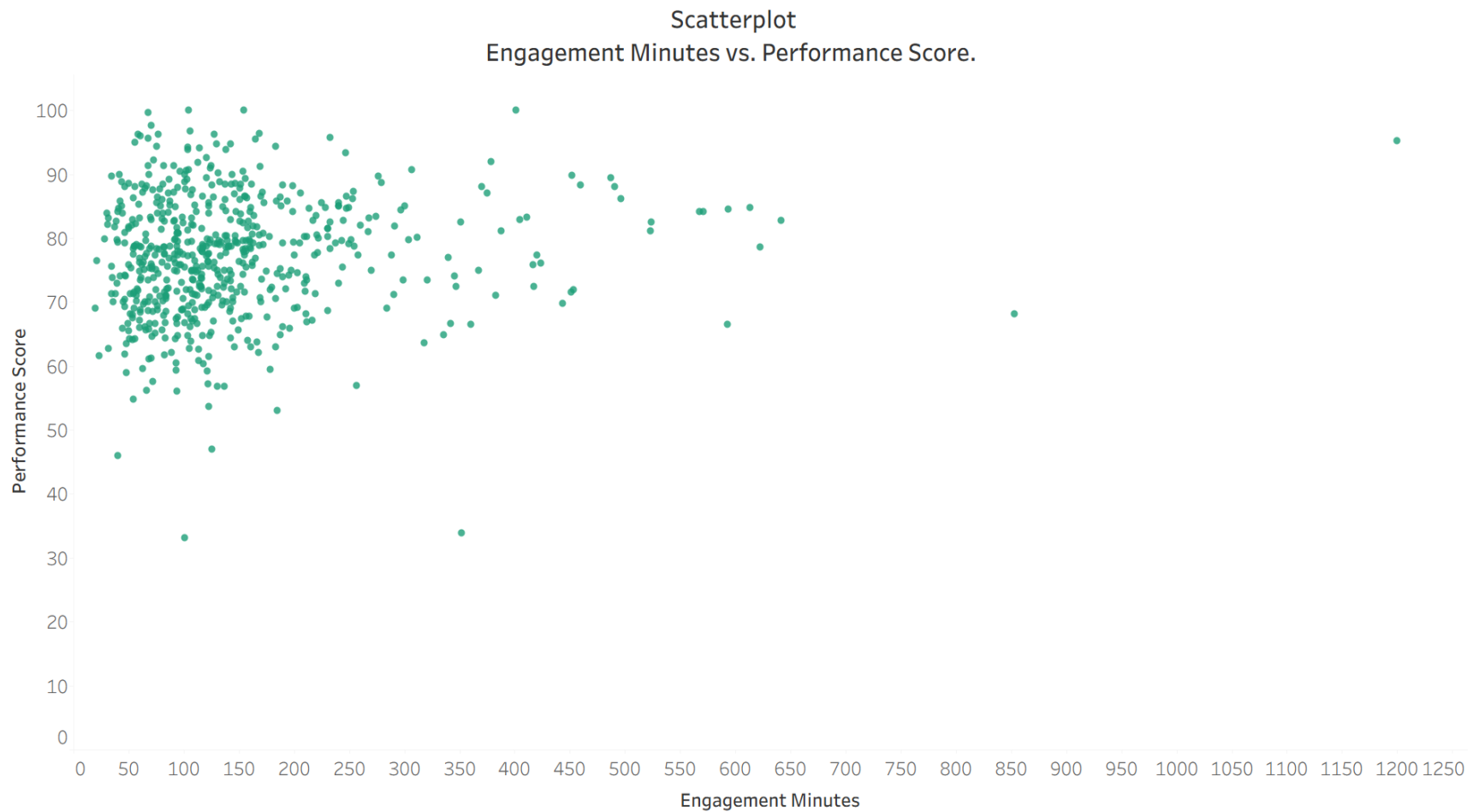
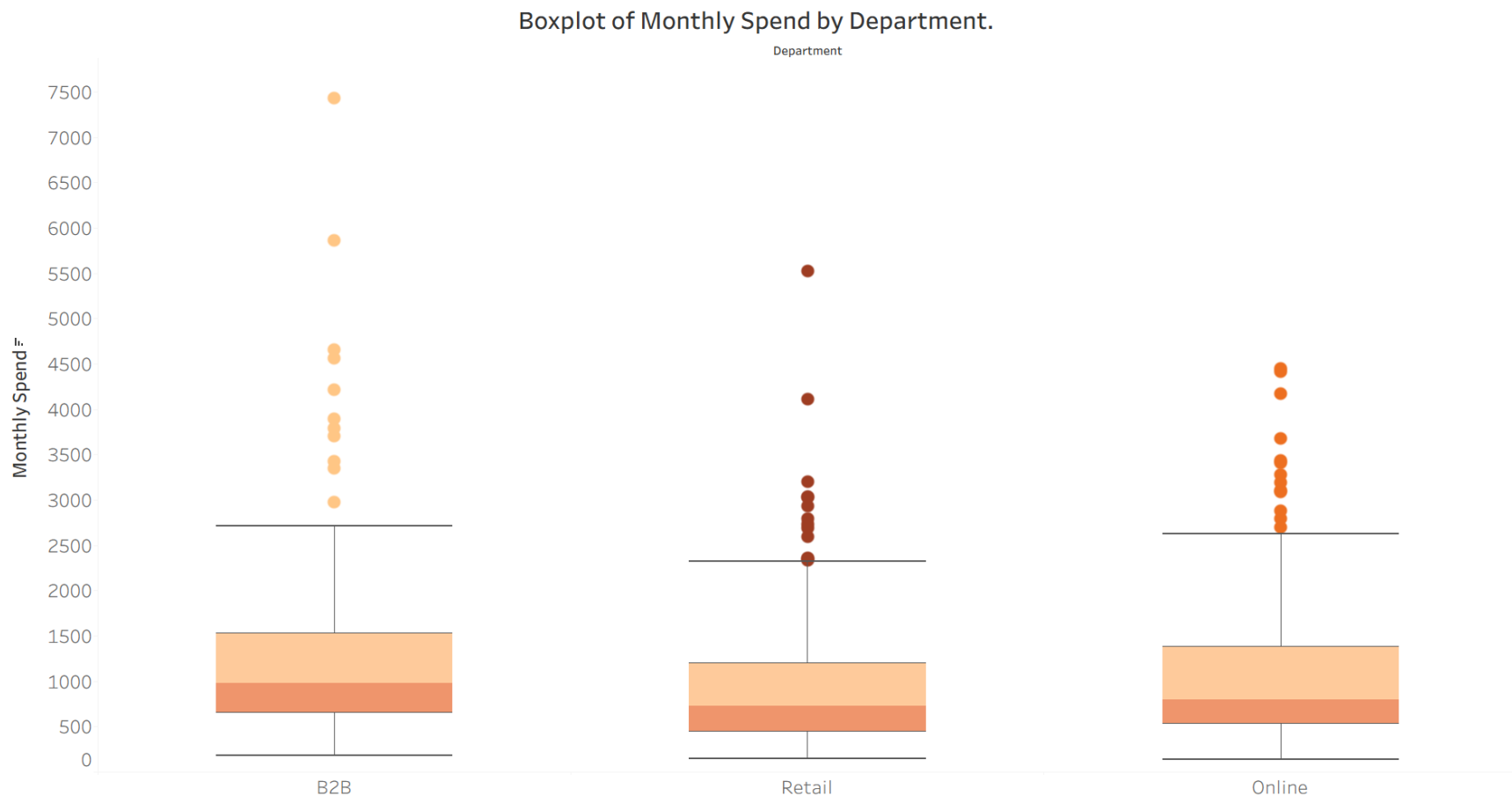


Figure B4

Monthly spending varies across departments, with B2B showing the highest median spend and the widest spread, indicating greater variability among business clients. Online customers display more outliers and a broader range, while Retail customers show more consistent spending patterns with fewer extreme values.



Appendix C Calculations

Calculation C1 IQR Outlier Detection

To illustrate the IQR outlier detection procedure, we use the variable **Age** as an example. The same steps are applied to all other continuous variables in the dataset.

Extract Quartiles from SPSS

From the *Descriptive* table for **Age**:

- **Q1 (25th percentile) = 31.75**
- **Q3 (75th percentile) = 47.00**

Compute the Interquartile Range (IQR)

$$IQR = Q3 - Q1 = 47 - 31.75 = 15.25$$

Compute the Outliers Bounds

Lower bound:

$$Q1 - 1.5 \times IQR = 31.75 - 1.5(15.25) = 8.875$$

Upper bound:

$$Q3 + 1.5 \times IQR = 47 + 1.5(15.25) = 69.875$$

Identify Outliers

Any case where:

- **Age < 8.875, or**

- **Age > 69.875**

is classified as an outlier.

Using the dataset (N = 600), SPSS shows **2 cases** above the upper bound

Age_Outlier					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	.00	598	99.7	99.7	99.7
	1.00	2	.3	.3	100.0
Total		600	100.0	100.0	

Compute the Outlier Percentage

$$Outlier \% = \frac{2}{600} \times 100 = 0.33 \%$$

Thus, **Age has 0.33% outliers**, which is below the 5% threshold.

Calculation C2 Z-score Outlier Detection

This calculation illustrates how Z-scores outliers were identified for the continuous variables.

The example below uses **Age**, and the same procedure is applied to all six continuous variables.

Generate Z-scores in SPSS

Analyze -> Descriptive Statistics -> Descriptives

Check the box: **“Save standardized values as variables”**

_Outlier	ZAge	
.00	-1.84199	
.00	-.19115	
.00	-.27804	
.00	-.62558	
.00	-.62558	
.00	-1.84199	
.00	-1.23378	
.00	2.50232	
.00	.41705	
.00	-.45181	
.00	-.79935	
.00	-1.06001	

Define the Outlier Rule

A case is considered an outlier if:

$$Z < -3 \text{ or } Z > 3$$

This is the standard Z-score outlier criterion.

Create an Outlier Indicator Using Compute Variable

$$(ZAge < -3) \text{ OR } (ZAge > 3)$$

Count the Number of Outliers

Age_Z_Outlier				
		Frequency	Percent	Cumulative Percent
Valid	.00	600	100.0	100.0

Compute the Outlier Percentage

$$\text{Outlier \%} = \frac{\text{Number of outliers}}{600} \times 100 = \frac{0}{600} \times 100 = 0.00\%$$

Using the ± 3 SD Z-score criterion, Age showed no values beyond -3 or 3, resulting in an outlier percentage of 0.00%. This confirms that no Z-score outliers were detected for Age, and the variable was retained in full for further analysis.

Calculation C3

Using the variable Monthly_Spend, which is approximately right skewed but large enough ($n = 600$) for the Central Limit Theorem to apply, we compute the probability that a randomly selected customer spends **more than \$2,000 per month**.

- Mean (μ) = **1083.86**
- Standard deviation (σ) = **868.28**

Compute Z-score

$$Z = \frac{2000 - \mu}{\sigma}$$

$$Z = \frac{2000 - 1083.86}{868.28} = 1.055$$

Find Probability

$$P(X > 2000) = 1 - \Phi(1.055)$$

Where $\Phi(1.055)$ represents the area to the left under the z-score 1.055

$$P(X > 2000) \approx 1 - 0.854 = 0.146$$

Calculation C4

Estimate the success probability p

A “success” is defined as selecting a customer from the **Retail** department.

$$p = \frac{207}{600} = 0.345$$

Let $X \sim \text{Binomial}(n = 10, p = 0.345)$.

We compute the probability of observing exactly four Retail customers in a sample of ten:

$$P(X = 4) = \binom{10}{4} (0.345)^4 (1 - 0.345)^6$$

$$P(X = 4) \approx 0.229$$

Appendix D Python Code

Theoretical Foundation of the Kruskal–Wallis Test

Note: This section is included *only to illustrate the theory behind the Kruskal–Wallis’s test*. It is **not part of the graded inferential analysis**.

The Kruskal–Wallis’s test is a non-parametric alternative to one-way ANOVA and is based on comparing the distributions of **pooled sample ranks** across groups. The hypotheses for the test are:

H_0 : All groups come from the same population distribution.

H_1 : At least one group comes from a different population distribution.

Under H_0 , when all observations from all groups are **pooled and ranked together**, the ranks follow a discrete uniform distribution. Kruskal and Wallis (1952) demonstrated that if all groups originate from the same continuous population, each observation is equally likely to receive any rank from 1 to n :

$$R_i \sim \text{Discrete Uniform}\{1, 2, \dots, n\}, P(R_i = k) = \frac{1}{n}$$

This uniformity forms the theoretical basis for comparing **mean ranks** across groups. If the groups differ, their rank distributions shift systematically higher-valued groups accumulate higher ranks, and lower-valued groups accumulate lower ranks. The Kruskal–Wallis’s statistic detects these systematic deviations from the uniform expectation.

Python Simulation of the Theoretical Rank Distribution

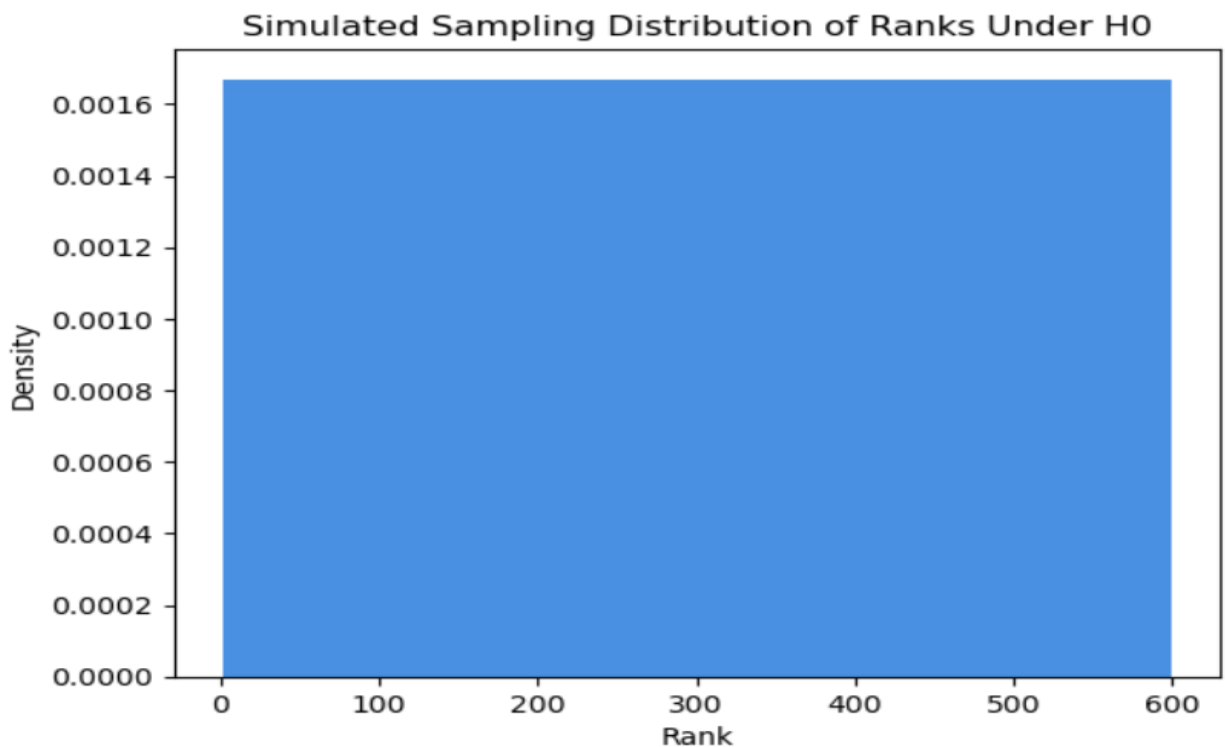
To illustrate this theoretical result, a simulation was conducted in Python by repeatedly generating random samples and computing their ranks. As expected, the sampling distribution of the ranks is perfectly uniform:

```
import numpy as np
import matplotlib.pyplot as plt

n = len(df)
simulated_ranks = []

for _ in range(10000):
    sample = np.random.rand(n)
    ranks = sample.argsort().argsort() + 1
    simulated_ranks.extend(ranks)

plt.hist(simulated_ranks, bins=n, density=True, color="#4A90E2")
plt.title("Simulated Sampling Distribution of Ranks Under H0")
plt.xlabel("Rank")
plt.ylabel("Density")
plt.show()
```



This confirms the theoretical result that, under H_0 , the ranks follow a discrete uniform distribution.

Rank Distributions by Department

To connect the theory to the actual dataset, the raw Monthly Spend values were transformed into **pooled ranks**, and separate histograms were generated for each department:

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_style("whitegrid")

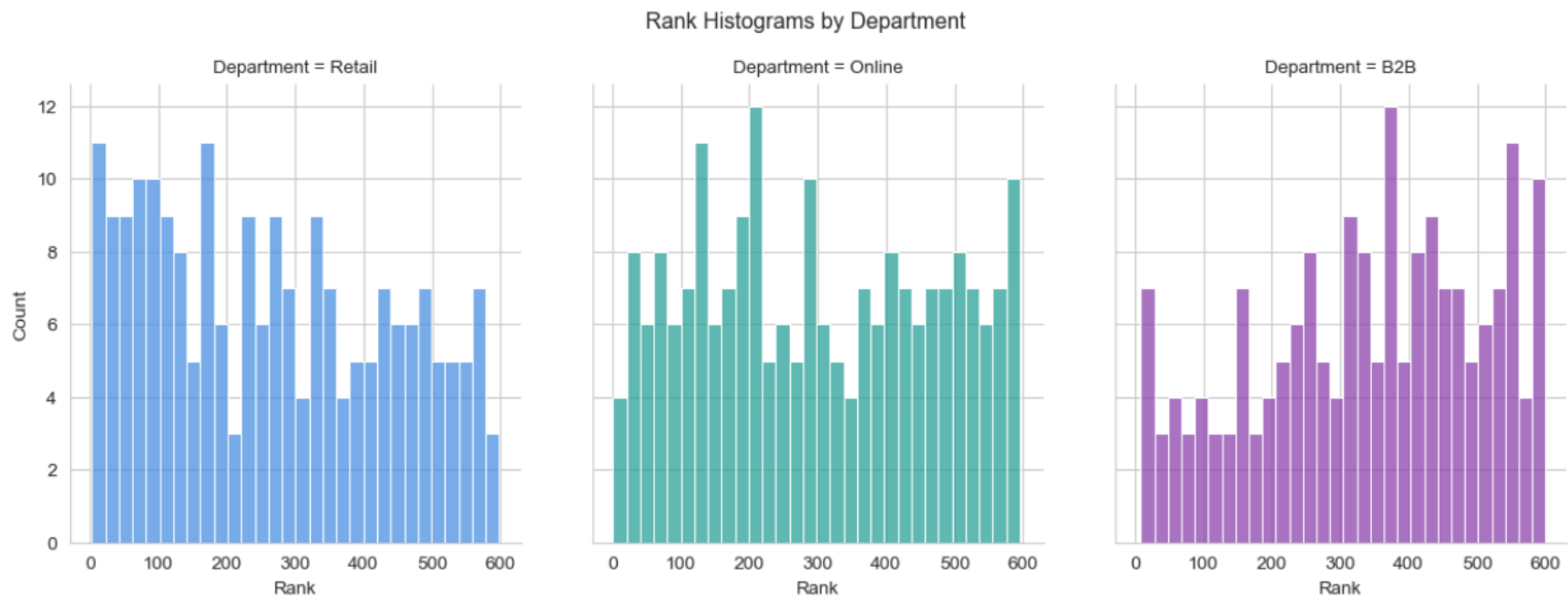
# Create pooled ranks
df['Rank'] = df['Monthly_Spend'].rank()

# Custom colors for each facet (Retail, Online, B2B)
facet_colors = ["#4A90E2", "#2AA198", "#8E44AD"] # blue, teal, purple

g = sns.FacetGrid(
    df,
    col="Department",
    col_wrap=3,
    height=4,
    sharex=True,
    sharey=True
)

# Apply a different color to each facet
for ax, color in zip(g.axes.flat, facet_colors):
    sns.histplot(
        df[df["Department"] == ax.get_title().split(" = ")[1]]["Rank"],
        bins=30,
        kde=False,
        color=color,
        alpha=0.75,
        ax=ax
    )

g.fig.suptitle("Rank Histograms by Department", y=1.05)
plt.show()
```



These plots show how each department's rank distribution shifts relative to the uniform expectation. Groups with higher spending accumulate **higher ranks**, while lower-spending groups cluster toward **lower ranks**. These shifts illustrate the logic behind the Kruskal–Wallis's test and visually support the inferential results.